WW-P-325B
June 9, 1976
SUPERSEDING
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April 5, 1967

FEDERAL SPECIFICATION

PIPE, BENDS, TRAPS, CAPS AND PLUGS; LEAD (FOR INDUSTRIAL PRESSURE, AND SOIL AND WASTE APPLICATIONS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lead pipe, bends, traps, caps and plugs.

1.2 Classification.

1.2.1 Pipe. The pipe shall be of the following classes, as specified (see 6.2):

Class:

100 - Industrial pressure pipe.
75 - Industrial pressure pipe.
50 - Industrial pressure pipe.
B - Soil and waste.
C - Soil and waste.
D - Soil and waste.

1.2.2 Traps. The traps shall be of the following types, as specified (see 6.2):

Type - P or 1/2 S.
Type - Drum.

1.2.3 Bends. The bends shall be of one class and type, as specified (see 6.2).

1.2.4 Sizes. The sizes shall be as specified (see 6.2).

FSC 4710, 4730

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650 - Crates, Wood, Open and Covered.

Federal Standards:

Fed Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC, 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation-Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

American National Standards Institute, Inc., (ANSI) Standard:

B2.1 - Pipe Threads (Except Dryseal).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM) Standards:

E37 - Chemical Analysis of Pig Lead.
E57 - Chemical Analysis of White Metal Bearing Alloys.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Society of Mechanical Engineers (ASME)

A40.8 - National Plumbing Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY, 10017.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent

Uniform Classification Committee.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard commercial product. The pipe, bends, traps, caps and plugs shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. A standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.2 Material.

3.2.1 Pipe, traps, and bends. The pipe, traps, and bends shall be of commercial lead containing not less than 99.7 percent lead, and the zinc content shall not exceed 0.002 percent (see 4.2.3.3).

3.2.2 Screw caps and plugs. Unless otherwise specified (see 6.2), caps and plugs shall be made of antimonial lead containing not less than 6 nor more than 10 percent antimony, not more than 0.002 percent zinc, and not more than 0.3 percent of elements other than lead and antimony. Screw caps and plugs shall have a Brinell hardness number of not less than 12 nor more than 15 (see 4.2.3.2 and 4.2.3.3).

3.2.3 Chemical analysis. Each melt shall be analyzed to insure compliance with 3.2.1 and 3.2.2. Results of each analysis shall be made available to the Government on request. When specified (see 6.2), samples from each lot of finished material shall be analyzed in accordance with 4.2.3.3.

3.3 Pipe. The pipe shall conform to the applicable dimensions and weight specified in Table I (see 4.2.2.3).

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TABLE I. Pipe size and weight.

Nominal size id (inches)	Commercial Designation (1) East	Commercial Designation (1) West	Nominal size od (inches)	Nominal wall (2) Thickness (inch)	Nominal weight per foot (pounds)
3/8	A	S	0.725	0.175	1.49
	AA	XS	.811	.218	2.00
	AAA	XXS	.888	.256	2.50
1/2	A	S	0.798	0.149	1.49
	AA	XS	.876	.188	1.99
	AAA	XXS	1.012	.256	2.99
5/8	A	S	1.019	0.197	2.55
	AA	XS	0.082	.228	3.01
	AAA	XXS	1.137	.256	3.48
3/4	A	S	1.156	0.203	2.99
	AA	XS	1.212	.231	3.50
	AAA	XXS	1.336	.293	4.72
1	A	S	1.43	0.214	4.01
	AA	XS	1.49	.246	4.73
	AAA	XXS	1.596	.298	5.97
1-1/4	D	XL	1.485	0.118	2.49
	C	L	1.53	.139	2.98
	B	M	1.59	.171	3.75
	A	S	1.67	.210	4.73
	AA	XS	1.765	.257	5.99
	AAA	XXS	1.89	.319	7.74
1-1/2	D	XL	1.775	0.138	3.49
	C	L	1.83	.165	4.24
	B	M	1.88	.191	4.98
	A	S	1.985	.242	6.51
	AA	XS	2.075	.288	7.95
	AAA	XXS	2.27	.386	11.24
1-3/4	D	XL	2.025	.137	3.99
	C	L	2.085	.168	4.97
	B	M	2.145	.198	5.95
	A	S	2.195	.221	6.74
	AA	XS	2.405	.327	10.5
	AAA	XXS	2.625	.437	14.8
2	D	XL	2.285	.142	4.69
	C	L	2.355	.177	5.94
	B	M	2.41	.205	6.97
	A	S	2.505	.251	8.74
	AA	XS	2.75	.375	13.8
	AAA	XXS	3.01	.504	19.4

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TABLE I. Pipe size and weight. (Con.)

Nominal size id (inches)	Commercial Designation (1) East	West	Nominal size od (inches)	Nominal wall (2) thickness (inch)	Nominal weight per foot (pounds)
2-1/2	D	XL	2.75	.125	5.07
	E	AQ	3.00	.250	10.63
3	C	L	3.25	.125	6.04
(3)			3.50	.250	12.56
4	C	L	4.25	.125	7.99
(3)			4.50	.250	16.42
5	--	--	5.25	.125	9.90
(3)			5.50	.250	20.29
6	--	--	6.25	.125	11.84
(3)			6.50	.250	24.15

(1) Note:

Class of pipe	Commerical designation East	West
100	AAA	XXS
75	AA	XS
50	A	XS
B	B	M
C	C	L
D	D	XL

(2) Specified thickness shall include a tolerance of ±0.010 inch for the wall thickness of the pipe.

(3) Pipe of 3-6 inches diameter is normally classified by weight per foot or by wall thickness.

3.3.1 Hydrostatic testing. Classes 100, 75, and 50 pressure pipe shall meet the hydrostatic requirements of 4.2.3.5.

3.4 Traps. The traps shall be either P or 1/2 S, or drum traps. The traps shall be of sufficient size to drain the fixture rapidly, but in no case shall the size be less than that indicated in the "Drainage System" requirements of ASME A40.8. The traps shall conform to the applicable dimensions and weights as specified in Table II and 3.4.2 (see 6.2).

TABLE II. Traps.

Nominal id (inches)	Weight per running foot (pounds)	Short traps Dimensions inlet (in.)	out (in.)	total (lb)	weight (oz)	Long traps Dimensions inlet (in.)	out (in.)	total (lb)	weight (oz)
1/2 S or P									
1-1/4	2-1/2	4-1/2	7	4	0	4-1/2	14	5	9
1-1/2	3	4-1/2	7	4	15	4-1/2	14	6	12
2	4	4-1/2	8	7	2	4-1/2	14	9	3

The seal for all 1/2 S and P traps measured vertically shall be not less than 2 and not more than 4 inches.

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3.4.1 Screw caps and plugs. All 1/2 S and P traps shall be equipped with a threaded cleanout plug so constructed as not to form a pocket sufficient to retain solid matter or capable of interfering with scouring operations. All threads shall be standard straight pipe threads conforming either to the requirements of ANSI B2.1 or of NBS H28.

3.4.2 Drum traps. Four-inch drum traps shall be made from pipe weighing 8 pounds per foot (lb/ft). Three-inch drum traps shall be made from pipe weighing 6 lb/ft. The nominal wall thickness shall be 0.125 inch and at no point shall be less than 0.118 inch. Drum traps are available in the following sizes: 3 by 8, 3 by 9, 3 by 10, 4 by 8, 4 by 9, and 4 by 10. Drum traps shall have a removable cover fitted in a threaded ring which shall be soldered, welded, or burned in place on the trap, to provide a tight joint. Each cover shall be furnished with a suitable washer of fiber or synthetic rubber. All threads shall be standard pipe threads conforming to the requirements of ANSI B2.1 or those of NBS H28. Each cover shall have either a raised square nut or countersunk wrench socket. Four-inch traps shall have two holes and 3-inch traps shall have one hole in the cover.

3.5 Bends. Bends shall conform to the dimensions and weights given in Table III. The most common ones are shown here (see 6.2).

TABLE III. Bends.

1-1/2 inches Nominal diameter			3 inches Nominal diameter			4 inches Nominal diameter		
Dimensions (1) (in.)	Total weight (2) (lb)	(oz)	Dimensions (1) (in.)	Total weight (2) (lb)	(oz)	Dimensions (1) (in.)	Total weight (2) (lb)	(oz)
4 by 7	2	12	5-1/2 by 10	7	3	5-1/2 by 10	9	5
4 by 12	4	1	5-1/2 by 12	8	3	5-1/2 by 12	10	10
4 by 15	4	13	5-1/2 by 15	9	12	5-1/2 by 15	12	10
4 by 18	5	10	5-1/2 by 18	11	4	5-1/2 by 18	14	10
4 by 20	6	2	5-1/2 by 20	12	4	5-1/2 by 20	16	0
7 by 7	3	8	10 by 10	9	8	10 by 10	12	5
7 by 12	4	13	10 by 12	10	8	10 by 12	13	10
7 by 15	5	10	10 by 15	12	0	10 by 15	15	10
7 by 18	6	6	10 by 18	13	8	10 by 18	17	10
7 by 20	6	15	10 by 20	14	8	10 by 20	18	15

(1) Measured from the center of the bend to the inlet and outlet ends, respectively.
(2) A minus tolerance of 5 percent is permitted. No maximum limitation.

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3.6 Marking for inspection.

3.6.1 Pipe. For sizes under 2-1/2 inches, the name of the manufacturer or his trademark, and the class and inside diameter (id) shall be stamped at least once on every length and coil of pipe. For sizes larger than 2-1/2 inches, the nominal weight per foot in pounds and id shall be stamped on each length of pipe.

3.6.2 Traps and bends. Each trap and bend shall be stamped with the name of the manufacturer or his trademark, and the id shall be indicated in inches.

3.6.3 Caps and plugs. Each cap and plug shall be stamped with the name of the manufacturer or his trademark, and the diameter shall be indicated in inches.

3.7 Workmanship. Workmanship shall be in accordance with the best commercial practice for this type of product. All pipe, bends, traps, caps and plugs shall be free of burrs, buckles, kinks, indentations, cracks, voids, pits, porosity, and other defects affecting the serviceability of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary.

4.2 Quality conformance inspection. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except when otherwise indicated. For purposes of sampling, an inspection lot shall consist of all material of the same class, size, and type of item submitted for delivery at one time.

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4.2.2 Examinations.

4.2.2.1 Inspection levels and Acceptable Quality Levels (AQL) for examinations. The inspection levels for determining the sample size and the AQL, expressed as percent defective, shall be as shown in Table IV.

TABLE IV. Inspection levels and AQL for examinations.

Examination paragraph	Inspection level	AQL
4.2.2.2	I	1.5
4.2.2.3	II	2.5

4.2.2.2 Examination for defects in appearance, construction, workmanship, and markings shall be as shown in Table V. The sample unit for this examination shall be one pipe, trap, bend, cap or plug.

TABLE V. Criteria for examination of defects.

Examine	Defects
Appearance, construction, and workmanship	Not the specified type of class. Component missing, damaged, defective, or not as specified. Not clean and smooth. Not sound (porous, cracked, or fractured). Threaded components have burred, or damaged threads, or are not as specified. Traps do not have holes as specified.
Markings	Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.2.2.3 Examination for dimensions and weight. Samples shall be selected from each lot in accordance with 4.2.2.1 and shall be measured and weighed to determine conformance with the requirements specified herein.

4.2.3 Tests.

4.2.3.1 Sampling for hardness test. A random sample of screw caps or plugs shall be taken from each lot in accordance with inspection level S-3 of MIL-STD-105 for hardness test specified in 4.2.3.2.

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4.2.3.2 Hardness. Each of the samples selected in accordance with 4.2.3.1 shall be subjected to a hardness test. The test shall be as specified in Method 242.1 of Fed. Test Method Std. No. 151. If any sample fails to meet the requirements in 3.2.2 the lot shall be rejected.

4.2.3.3 Chemical analysis. When specified (see 3.2.3), two samples shall be taken from each lot of pipe, traps, bends, caps and plugs for determination of chemical composition in accordance with ASTM E57, ASTM E37, or other analytical technique of equal sensitivity. If any sample fails to meet the requirements of 3.2.1 and 3.2.2, the lot shall be rejected.

4.2.3.4 Sampling for hydrostatic test. A random sample of industrial pressure pipe shall be taken from each lot in accordance with inspection level S-3 of MIL-STD-105 for the hydrostatic test specified in 4.2.3.5.

4.2.3.5 Hydrostatic test. Each of the samples selected in accordance with 4.2.3.4 shall be subjected to a hydrostatic pressure test for a period of 1 minute as applicable.

Class	Test pressure, psi
100	200
75	150
50	100

Samples shall show no leaks or defects as a result of this test.

4.2.4 Inspection of preparation for delivery requirements. The packing and marking of the pipe, traps, and bends shall be examined to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be levels A or C (see 6.2).

5.1.2 Level A. The pipe, bends, and traps shall be preserved and packaged in accordance with MIL-P-116, Method III. The unit package quantity shall be as specified (see 6.2).

5.1.3 Level C. The pipe, bends, and traps shall be preserved and packaged in accordance with the contractor's standard practice.

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5.2 Packing. Packing shall be levels A, B, or C as specified (see 6.2).

5.2.1 Level A. Pipe, bends, and traps shall be packed in boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or crates conforming to PPP-C-650. The contents shall be cushioned, blocked, and braced to prevent movement or damage.

5.2.2 Level B. Level B shall be the same as specified in level A except the containers shall be of the domestic type or class 1, as applicable.

5.2.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition, and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract each unit package, intermediate package, and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Classes 100, 75, and 50. These classes are intended for use in industrial applications requiring pressure pipe; however, lead pipe shall not be used for the distribution of drinking water.

6.1.2 Classes B, C, and D. These classes are intended for applications in soil and waste systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class designation of pipe and nominal id size required (see 1.2 and Table I).
(c) Type of trap and size required (see 1.2 and 3.4).
(d) Nominal diameter and dimensions of bends required (see 3.5).

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(e) When screw caps and plugs of other materials are required (see 3.2.2).
(f) When chemical analysis are required (see 3.2.3).
(g) Selection of applicable level of preservation and packaging and packing required (see 5.1 and 5.2).

6.3 The ASME A40.8 normally does not recommend the use of full S traps and they are not covered by this specification.

6.4 Pipe should be ordered by number of feet required in preference to weight and traps, and bends should be ordered by the piece.

6.5 The standard sizes of pipe, traps, and bends specified herein conform to the Commercial Standards for Lead Pipe and Lead Traps and Bends, respectively, CS 95-41 and CS 96-41 as developed by the trade in cooperation with the National Bureau of Standards.

6.6 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.7 Cross reference for classifications. The following changes were made to the classifications of pipe in this specification:

WW-P-325A	WW-P-325B
Class: 100-Service and supply	Class: 100-Industrial pressure pipe
75-Service and supply	75-Industrial pressure pipe
50-Service and supply	50-Industrial pressure pipe
B-Soil and waste	B-Soil and waste
C-Soil and waste	C-Soil and waste
D-Soil and waste	D-Soil and waste

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Military Custodians:

Navy - YD
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Review activity:

DCSC

Civil Agency Coordinating Activities:

GSA-FSS
HEW-FEC

Preparing activity:

Navy - YD

Project No. 4730-0254

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